

MODULE 1: OVERVIEW OF OPERATING SYSTEMS
WEEK 1

Learning Outcomes:

After completing this course you are expected to understand the following:

- 1. Role and purpose of the operating system
- 2. Functionality of a typical operating system
- 3. Mechanisms to support client-server models, hand-held devices
- 4. Design issues (efficiency, robustness, flexibility, portability, security, compatibility)
- 5. Influences of security, networking, multimedia, windowing systems

A. Engage

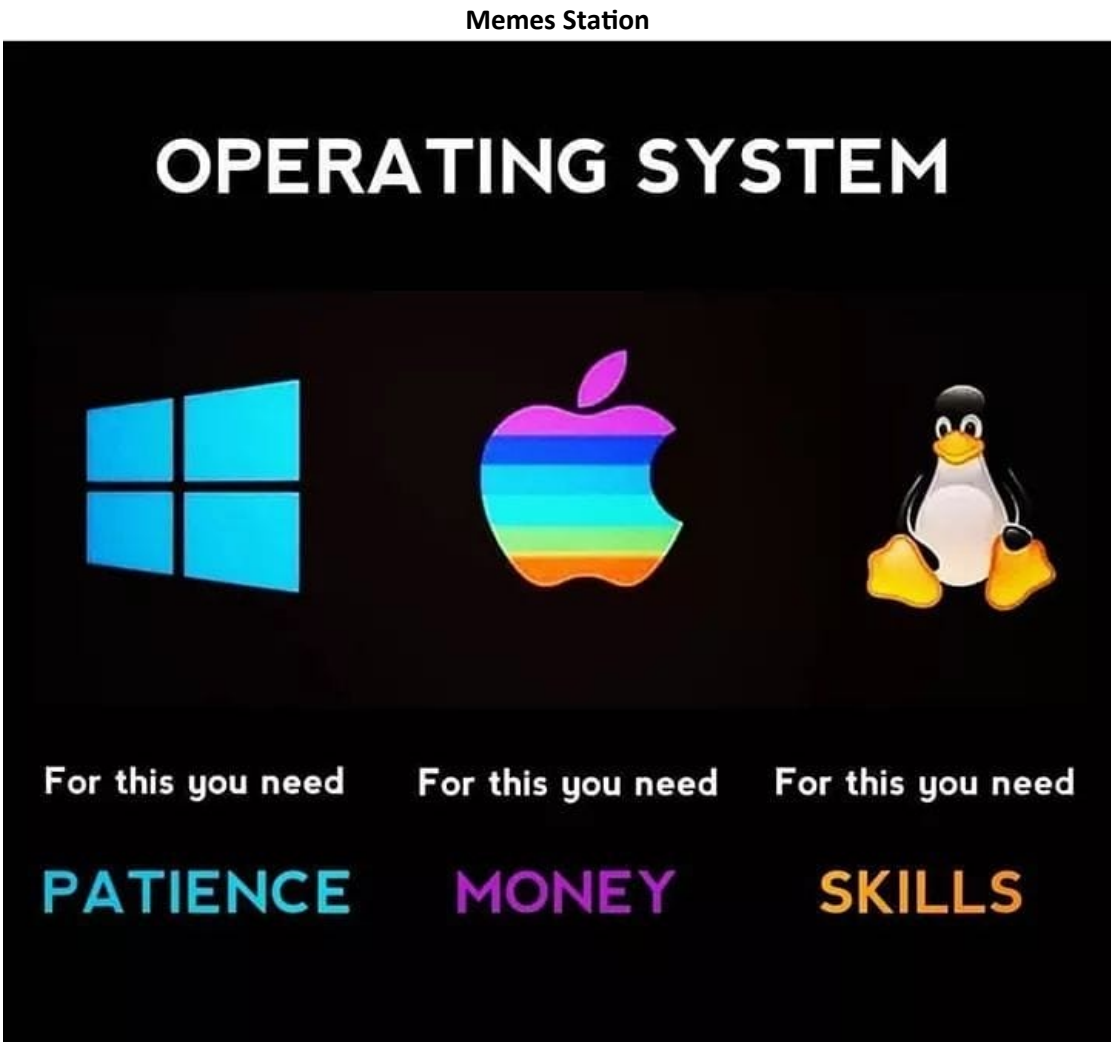


Figure 1.0: 3 COMMON OPERATING SYSTEM.

B. Explain

The operating system you use was most likely pre-loaded onto your computer long before it got to you. The three most common operating systems for personal computers are Microsoft Windows, macOS, and Linux. Operating systems use a graphical user interface, or GUI (pronounced gooey), that lets your mouse click buttons, icons, and menus, and displays graphics and text clearly on your screen.

Microsoft created the Windows operating system in the mid-1980s and it is the most popular OS in the world. Windows comes pre-loaded on most new personal computers (PCs).

C. Explore

Video Titles:

1. What is an Operating System as Fast As Possible

YouTube Links:

1. <https://www.youtube.com/watch?v=pVzRTmdd9j0>

Module Video Filenames:

1. WEEK01_Operating System

D. Elaborate

Introduction

1. Role and purpose of the operating system

In the Computer System (comprises of Hardware and software), Hardware can only understand machine code (in the form of 0 and 1) which doesn't make any sense to a naive user.

We need a system which can act as an intermediary and manage all the processes and resources present in the system.

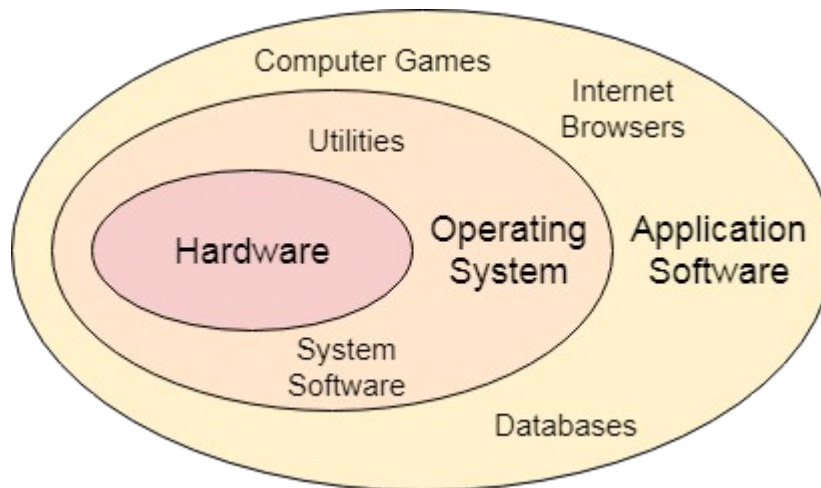


Figure 1.1. Operating System Ecosystem

2. Functionality of a typical operating system.

An **Operating System** can be defined as an **interface between user and hardware**. It is responsible for the execution of all the processes, Resource Allocation, CPU management, File Management and many other tasks.

The purpose of an operating system is to provide an environment in which a user can execute programs in convenient and efficient manner.

Structure of a Computer System

A Computer System consists of:

- Users (people who are using the computer)
- Application Programs (Compilers, Databases, Games, Video player, Browsers, etc.)
- System Programs (Shells, Editors, Compilers, etc.)
- Operating System (A special program which acts as an interface between user and hardware)

- Hardware (CPU, Disks, Memory, etc).

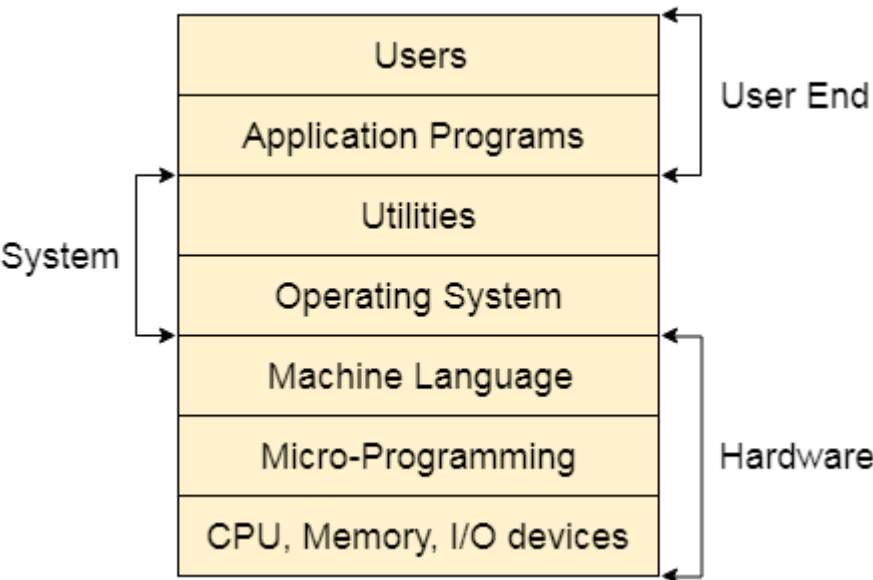


Figure 1.2. Structure of a Computer System

3. Mechanisms to support client-server models, hand-held devices.

Client-Server network operating system: It is the type of network operating system that allows the users to access resources, functions, and applications through a common server or center hub of the resources. The client workstation can access all resources that exist in the central hub of the network. Multiple clients can access and share different types of the resource over the network from different locations.

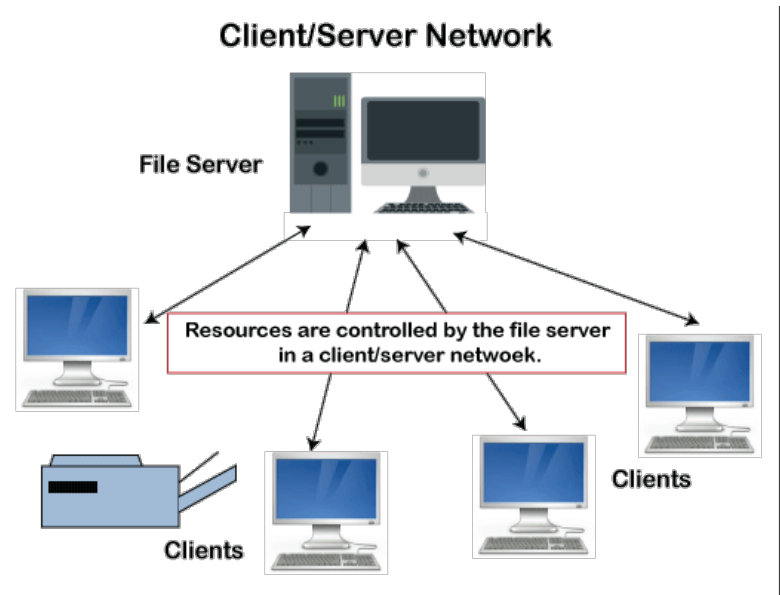


Figure 1.3. Client/Server Model

Mobile OS

A mobile operating system is an operating system for mobile phones, tablets, smartwatches, 2-in-1 PCs, smart speakers, or other mobile devices. While computers such as typical laptops are 'mobile', the operating systems used on them are generally not considered mobile ones, as they were originally designed for desktop computers that historically did not have or need specific *mobile* features. This distinction is becoming blurred in some newer operating systems that are hybrids made for both uses.

Mobile devices, with mobile communications abilities, contain two mobile operating systems – the main user-facing software platform is supplemented by a second low-level proprietary real-time operating system which operates the radio and other hardware. Research has shown that these low-level systems may contain a range of security vulnerabilities permitting malicious base stations to gain high levels of control over the mobile device.

4. Design issues. Influences of security, networking, multimedia, windowing systems.

Advantage of using Operating System

- Allows you to hide details of hardware by creating an abstraction
- Easy to use with a GUI
- Offers an environment in which a user may execute programs/applications
- The operating system must make sure that the computer system convenient to use
- Operating System acts as an intermediary among applications and the hardware components
- It provides the computer system resources with easy to use format
- Acts as an intermediary between all hardware's and software's of the system

Disadvantages of using Operating System

- If any issue occurs in OS, you may lose all the contents which have been stored in your system
- Operating system's software is quite expensive for small size organization which adds burden on them. Example Windows
- It is never entirely secure as a threat can occur at any time

E. Evaluate

ASSESSMENT:

Instructions: You may write your answer on the Answer Sheet (AS) provided on this module.

I. TRUE OR FALSE (2-Point Each):

1. Client-Server is the type of network operating system that allows the users to access resources through a common server or center hub of the resources.
2. The purpose of an operating system is to provide an environment in which a user can execute programs in convenient and efficient manner.
3. Software can only understand machine code (in the form of 0 and 1) which doesn't make any sense to a naive user.
4. An Operating System can be defined as an interface between user and hardware.
5. CPU is responsible for the execution of all the processes, Resource Allocation, CPU management, File Management and many other tasks.
6. The purpose of an operating system is to provide an environment in which a user can execute programs in convenient and efficient manner.
7. Multiple clients can access and share different types of the resource over the network from different locations.
8. Mobile devices contain two mobile operating systems – the main user-facing software platform and real-time operating system.
9. Operating System acts as an intermediary among applications and the hardware components
10. If any issue occurs in OS, you may backup all the contents which have been stored in your system

References:

- 1. History of the Operating System–
<https://www.javatpoint.com/history-of-operating-system>
- 2. Operating System - Services–
https://www.tutorialspoint.com/operating_system/os_services.htm

Facilitated By:		
Name	:	
MS Teams Account (email)	:	
Facebook Page	:	